A Fixed Iterated Greedy Sequence Cuts Weighted Tardiness on an Eight-Stage Flow Line by More than 2,500 Weighted Minutes per Period at Every Level of Processing-Time Variation Tested

OM 522 Team Project, Option 6

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Abstract. On a flow line where jobs cannot pass one another, a production planner must commit to one job order for every stage, and the earliest due date rule ignores job weights, release times, and the processing times at later stages. We applied the iterated greedy algorithm that Ruiz and Stützle proposed for flow shops with sequence-dependent setup times to an eight-stage line without setups but with release times and weighted due dates, and compared its fixed sequence with the earliest due date sequence on exact data and on 180 paired scenarios of random processing times. On development instances with 15, 45, and 120 jobs, iterated greedy lowered total weighted tardiness by 98.5, 97.6, and 81.0 percent, and a mixed-integer program solved with HiGHS proved its values optimal on the 6-job and 15-job instances. When each processing time of the 45-job instance varied with a coefficient of variation of 0.2, the iterated greedy plan had the lower weighted tardiness in all 30 scenarios, with a mean of 420.6 weighted minutes against 3,541.1, an 88.1 percent reduction. As the coefficient of variation rose from 0.1 to 0.8, the relative reduction fell from 94.6 to 27.2 percent while the mean absolute reduction stayed between 2,572.0 and 3,683.7 weighted minutes per period, and at 0.8 the 10th percentile of the per-period reduction fell to 851.5 and the earliest due date plan won 1 of 30 scenarios. A planner can therefore adopt the searched sequence for its absolute gain and, on the evidence of the one factor we varied, should expect that gain to arrive less dependably in a single period as processing times grow less predictable.

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1. Background

On a flow line such as printed circuit board assembly or furniture finishing, every job visits the same stages in the same order, and when jobs cannot pass one another, the order chosen at the first stage holds at every stage. On the line studied here operations take 2 to 20 minutes, so a job that needs 2 minutes at the first stage and 20 at a later one would delay every job behind it at that later stage, and the waiting would build up downstream. The planner chooses one sequence for a period’s jobs, which are released at different times and carry due dates of different importance, expressed as weights. Vallada, Ruiz, and Minella report that dispatching rules are used very often in practice and that earliest due date is among the most common rules for total tardiness [1], and the course project makes it the rule that a researched method must beat. Earliest due date orders the jobs by due date alone, so it cannot see a job’s weight, its release time, or its processing times at the stages after the first.

Minimizing total tardiness on a permutation flow shop is NP-hard in the ordinary sense with one machine and in the strong sense with two or more [1], and on this line an exact solver needed 85.8 seconds to certify the optimum for only 15 jobs, so a planner sequencing 45 to 120 jobs per period needs a heuristic. Ruiz and Stützle’s iterated greedy algorithm repeatedly removes a few jobs from a sequence and reinserts each at its best position [2], and on flow shops with sequence-dependent setup times and the weighted tardiness objective it had the lowest average deviation from the best known solutions among the methods they tested [3]. In a review of 40 methods for total tardiness, the heuristics built on inserting and interchanging jobs performed best among the heuristics, ahead of every dispatching rule [1]. These evaluations used instances with fixed processing times and no release times, and none of them executed a searched sequence on processing times that differ from the plan. A sequence tuned to the exact times in a data file could lose its advantage once the 2- to 20-minute operations run long or short, and a planner who must fix the order in advance cannot tell from these studies whether it does.

We implemented the earliest due date rule exactly as the course project states it and applied iterated greedy with local search [3] to an eight-stage line with release times and no setup times. A mixed-integer program solved with HiGHS [4] certified the optimal values of the two smallest instances, and a checker that shares no code with either method scored every schedule. We then held each method’s sequence fixed and executed it on 30 scenarios in which every processing time was multiplied by an independent lognormal factor with mean 1 and a coefficient of variation of 0.2, and on 30 further scenarios at each of five levels of variation from 0.1 to 0.8.

On exact data, iterated greedy left 1.5, 2.4, and 19.0 percent of the rule’s weighted tardiness on the instances with 15, 45, and 120 jobs, and it matched the proven optimum on the 6-job and 15-job instances. Held fixed on the 45-job instance at a coefficient of variation of 0.2, its plan left 11.5 jobs late per period on average, against 36.1 for the rule’s plan, and across all six sets of scenarios it had the lower weighted tardiness in 179 of 180. The mean reduction stayed between 2,572.0 and 3,683.7 weighted minutes at every level of variation, while the 10th percentile of the reduction in a single period fell from at least 1,781.1 weighted minutes at coefficients up to 0.5 to 851.5 at 0.8.

A planner can adopt the searched sequence for an absolute gain that held at every level of variation we

tested, at the cost of a sequence that cannot be traced by hand and seconds of computation. Processing-time variability is the one condition we varied, and because it eroded the gain in the worst periods first, it is the first quantity to measure on a line before relying on the plan.

2. Data

The data are synthetic instances that the course instructor generated for option 6 of the OM 522 team project, with all times in minutes. Each instance consists of a file of jobs, giving each job's release time, due date, and weight, a file of stages, giving their order on the line, and a file with one positive processing time for every job and stage. Table 1 summarizes the five instances: a 6-job case on 4 stages meant for tracing by hand, development instances with 15, 45, and 120 jobs on 8 stages, and a 10-job edge case on 4 stages.

Table 1: The five supplied instances. Processing times are per job and stage. The due-date window is the time from a job's release to its due date, and its median is taken over the instance's jobs. All times are in minutes.

Instance	Jobs	Stages	Processing times	Releases	Median due-date window	Weights
Hand-checkable	6	4	3 to 19	8 to 78	30	1, 2, 4
Small development	15	8	2 to 20	24 to 210	158	1, 2, 4, 5
Medium development	45	8	2 to 20	10 to 607	154	1 to 5
Large development	120	8	2 to 20	3 to 1,690	154	1 to 5
Edge case	10	4	0 to 20	5 to 126	87.5	1, 2, 3, 5

The three development instances share one generator, so their processing times range from 2 to 20 minutes and average between 85.1 and 88.8 minutes of total processing per job over the 8 stages. Their releases spread over 186, 597, and 1,687 minutes, which gives arrival rates between 0.071 and 0.081 jobs per minute, so the larger instances cover longer planning periods at a similar density. The median time from release to due date is 154 to 158 minutes, and subtracting each job's own total processing time leaves a median of 64.5 to 74 minutes that a job can spend waiting before it is late. Weights are integer priority scores from 1 to 5, and a weighted minute is one minute of lateness multiplied by the job's priority.

Before solving, our tool applies the four input checks the course project requires: every required file and column is present, every value has its declared type, every job and stage referenced by a processing time exists, and every processing time is positive. It reports each failure with its file, column, and line, and it never replaces a missing or invalid value. The edge case contains a processing time of zero, so the tool rejects it without writing a decision, as the course project specifies for this option.

3. Objective

A job's tardiness is the time by which its completion at the last stage exceeds its due date, and it is zero when the job finishes on time. The primary measure is total weighted tardiness, the sum over jobs of weight times tardiness, in weighted minutes. Due dates are soft targets, so a late job makes a schedule worse without making it infeasible.

Total waiting sums, over jobs and over every stage after the first, the time from a job's completion at one stage to its start at the next. The course project allows unlimited waiting space between stages, so

waiting measures work in process piling up on the line and carries no feasibility consequence. Every comparison also reports the computation time of each method.

Iterated greedy compares sequences by total weighted tardiness and breaks exact ties by total waiting, because many sequences can share a tardiness value when most jobs finish on time. The earliest due date rule optimizes nothing; it sorts the jobs once. The mixed-integer program minimizes total weighted tardiness alone. Every reported value of either measure is recomputed after the fact from the exported schedule, so no method reports its own score.

4. Model formulation

4.1 One sequence fixes every start time

The line processes n jobs, indexed by j, on m stages, indexed by k in line order. Job j is released at time r_j, is due at d_j, carries weight w_j, and needs processing time $p_{jk} > 0$ at stage k. A sequence π lists the jobs in the order they enter the line, with $\pi(i)$ the job in position i, and the same order holds at every stage. Given π, each operation starts at the later of the job's completion at the previous stage and the stage's completion of the previous job:

$$S_{\pi(i),k} = \max\bigl\{C_{\pi(i),k-1},\ C_{\pi(i-1),k}\bigr\}, \qquad C_{\pi(i),k} = S_{\pi(i),k} + p_{\pi(i),k}, \tag{4.1}$$

with the conventions $C_{\pi(i),0} = r_{\pi(i)}$, so the first stage waits for the release, and $C_{\pi(0),k} = -\infty$, so the first job waits for no other.

The tardiness of job j is $T_j = \max\{0,\ C_{jm} - d_j\}$, and the two measures of section 3 are

$$Z(\pi) = \sum_{j=1}^{n} w_j T_j, \qquad W(\pi) = \sum_{j=1}^{n} \sum_{k=2}^{m} \bigl(S_{jk} - C_{j,k-1}\bigr). \tag{4.2}$$

By induction over positions and stages, any schedule that keeps sequence π and satisfies the constraints of the course project starts each operation no earlier than (4.1), so the recursion gives every job its earliest completion at the last stage and π its smallest value of Z. The planning problem is therefore to choose the sequence π that minimizes $Z(\pi)$, and (4.1) is also the execution rule that the course project prescribes for the stress test.

4.2 A mixed-integer program defines the optimum

To certify optimal values, we state the problem as a mixed-integer program over sequence positions $q = 1, \dots, n$. The binary variable x_{jq} equals 1 when job j takes position q, B_{qk} and E_{qk} are the start and

end of the job in position q at stage k, and $T_j \geq 0$ is the tardiness of job j:

$$\min \quad \sum_{j=1}^{n} w_j T_j \tag{4.3}$$

$$\text{s.t.} \quad \sum_{q=1}^{n} x_{jq} = 1 \quad \forall j, \qquad \sum_{j=1}^{n} x_{jq} = 1 \quad \forall q, \tag{4.4}$$

$$E_{qk} = B_{qk} + \sum_{j=1}^{n} p_{jk}\, x_{jq} \quad \forall q, k, \tag{4.5}$$

$$B_{q1} \geq \sum_{j=1}^{n} r_j\, x_{jq} \quad \forall q, \tag{4.6}$$

$$B_{qk} \geq E_{q,k-1} \quad \forall q,\ k \geq 2, \tag{4.7}$$

$$B_{qk} \geq E_{q-1,k} \quad \forall q \geq 2,\ k, \tag{4.8}$$

$$T_j \geq E_{qm} - d_j - U_j\,(1 - x_{jq}) \quad \forall j, q, \tag{4.9}$$

$$x_{jq} \in \{0, 1\}, \quad T_j \geq 0.$$

Constraints (4.4) place every job in exactly one position, (4.5) runs each operation without interruption, (4.6) and (4.7) enforce release and stage order, and (4.8) lets a stage start a job only after finishing the previous position, which also keeps one order at every stage. Constraint (4.9) charges the tardiness of the job in position q to that job, and it is inactive for every other job when $U_j = \max\{0,\ \max_h r_h + \sum_{h,k} p_{hk} - d_j\}$, with h running over all jobs.

Under recursion (4.1), every job of every sequence finishes by $\max_h r_h + \sum_{h,k} p_{hk}$, because a completion time is a release plus processing times along one path through the schedule. The bound U_j therefore never forces a positive tardiness on the earliest-start schedule of any sequence, which by section 4.1 is the best schedule for that sequence. The program's optimal value equals $\min_\pi Z(\pi)$, and we use it only as a benchmark, never as one of the two compared methods.

4.3 The rule sorts once, and iterated greedy searches by insertion

The earliest due date rule sorts the jobs by d_j, breaks ties by job identifier, and times the resulting sequence by (4.1), exactly as the course project defines it.

Our researched method is IG_RS_LS, the iterated greedy algorithm with local search that Ruiz and Stützle proposed for the flow shop with sequence-dependent setup times and the makespan or weighted tardiness objective [3]; our line has no setup times, so we apply its weighted tardiness version with every setup time equal to zero, as shown in Algorithm 1. It compares sequences by the pair $F(\pi) = (Z(\pi), W(\pi))$ in lexicographic order, and the best position for a job is the position in a partial sequence that gives the smallest F, the earliest one on an exact tie. The start, NEH_EWDD, orders the jobs by ascending d_j / w_j and inserts them one at a time at their best positions [3], following the insertion steps of NEH as stated by Ruiz and Stützle [2]. Each iteration removes $\delta = 4$ jobs at random, reinserts them in the order removed, and applies a local search that moves single jobs to their best positions until no move improves F. A worse sequence replaces the current one with probability $\exp(-\Delta/\Theta)$, where Δ is its increase in Z and $\Theta = \lambda \sum_{j,k} p_{jk} / (10\, n\, m)$ with $\lambda = 0.5$, the constant-temperature rule and settings of the source [3].

We break ties in Z by waiting, because the course project names waiting as the secondary measure and the source compares on one objective. We start from the earliest due date sequence whenever it beats NEH_EWDD, so the method can never return a sequence worse than the rule; on all four instances NEH_EWDD was the better start. We stop after $L = 5000$ iterations with a fixed seed of 522, where the

Algorithm 1 Iterated greedy for weighted tardiness, after Ruiz and Stützle [3]. $F = (Z, W)$ is compared lexicographically, and every insertion goes to the best position.

Require: instance; destruction size $\delta = 4$; temperature factor $\lambda = 0.5$; iteration limit $L = 5000$; seed 522
$\pi \leftarrow$ NEH_EWDD: jobs by ascending d_j / w_j, each inserted at its best position
if $F(\pi^{\mathrm{EDD}}) < F(\pi)$ **then** $\pi \leftarrow \pi^{\mathrm{EDD}}$ ▷ never start worse than the rule
end if
$\pi \leftarrow$ LocalSearch(π); $\pi^{*} \leftarrow \pi$; $\Theta \leftarrow \lambda \sum_{j,k} p_{jk} / (10\, n\, m)$
for $\ell = 1, \ldots, L$ **do**
  remove δ jobs chosen at random from π, keeping the removed jobs in order of removal
  insert each removed job, in that order, at its best position
  $\pi' \leftarrow$ LocalSearch of the rebuilt sequence
  if $F(\pi') < F(\pi)$ **then** $\pi \leftarrow \pi'$; **if** $F(\pi) < F(\pi^{*})$ **then** $\pi^{*} \leftarrow \pi$
  else if $u \le \exp\bigl(-(Z(\pi') - Z(\pi))/\Theta\bigr)$ for u uniform on $[0, 1]$ **then** $\pi \leftarrow \pi'$
  end if
end for
return π^{*}

function LocalSearch(π)
  repeat
    for each job j, in random order **do**
      remove j and insert it at its best position, giving π''
      if $F(\pi'') < F(\pi)$ **then** $\pi \leftarrow \pi''$
      end if
    end for
  until a full pass changes nothing
  return π
end function

source stops at a CPU-time limit, because the course requires a rerun on another computer to reproduce the submitted result. The best-position search times the jobs ahead of each position once and abandons a position as soon as its running pair (Z, W) can no longer beat the best position found, which is safe because Z and W only grow as jobs are appended.

5. Experiment

5.1 Both methods ran on the same instances with fixed settings

Both methods ran on the hand-checkable instance and the three development instances, with the same input files and the measure definitions of section 3. Iterated greedy used its default settings of $L = 5000$ iterations and seed 522, with a safety cap of 300 seconds that no run reached, and we repeated it with seeds 1 to 5 to see how much the result depends on the seed. NEH_EWDD alone, the first step of Algorithm 1, appears in the comparison as a construction-only reference, so the share of the improvement due to the search can be read off. All runs used one laptop with an Intel Core Ultra 7 155U processor under Linux, Python 3.12, NumPy 2.5.4, and Numba 0.68.0, and the computation time covers the method alone, after one warm-up run that compiles the timing code. We expected iterated greedy to have the lower total weighted tardiness on every instance, because it evaluates the actual tardiness of each candidate sequence while the rule never does.

Each method's sequence was timed by (4.1) and exported as a list of start and completion times for every job and stage. A separate checker, written without any code from the methods, read only the input

files and the exported schedule, confirmed each constraint of the course project within a tolerance of 10^{-8} minutes, and recomputed Z, W, and the share of jobs finishing on time. Every number we report for a schedule comes from this checker.

5.2 HiGHS certified optimal values on the two smallest instances

We solved program (4.3) to (4.9) with HiGHS 1.15.1 [4] on the 6-job and 15-job instances, using one thread, a relative gap tolerance of zero, and a limit of 600 seconds. The job order of each solution was re-timed by (4.1) and confirmed by the checker, which supplied its Z and W. We did not run the program on the larger instances, because the 15-job instance already needed 85.8 seconds and more than 15,000 branch-and-bound nodes.

5.3 Thirty saved scenarios tested the fixed plans

The uncertain input is processing time, as the course project specifies for this option. In scenario ω, each processing time becomes $p_{jk}\,\varepsilon_{jk\omega}$, where the factors $\varepsilon_{jk\omega}$ are independent lognormal draws with mean 1 and coefficient of variation ν, obtained with $\sigma^2 = \ln(1+\nu^2)$ and log-mean $-\sigma^2/2$. For the main test we drew 30 scenarios at $\nu = 0.2$ for the 45-job instance from seed 20261103 and saved them to files before either method ran, so both plans faced identical realizations. Each method's plan is its sequence for the exact data under default settings. Each plan kept its sequence in every scenario and was timed by (4.1) with the realized times, the course project's execution rule, and the checker confirmed each realized schedule against that scenario's times.

For each plan we report the mean and the 90th percentile, by linear interpolation, of Z and W over the 30 scenarios, and the number of scenarios in which the plan breaks a hard constraint. To find where the comparison changes, we repeated the execution with 30 new scenarios at each of $\nu = 0.1$, 0.2, 0.3, 0.5, and 0.8, drawn from seeds 6522 to 6526. We expected both plans to grow later as ν grows and the iterated greedy plan to lose part of its advantage, because a plan tuned to exact times has more on-time jobs that a delay can make late.

6. Results

6.1 Iterated greedy lowers weighted tardiness on every instance and reaches the proven optima

Table 2 reports every method on every instance, and every one of these schedules is feasible. Iterated greedy lowered total weighted tardiness relative to earliest due date by 17.1 percent on the 6-job instance, from 281 to 233 weighted minutes, and by 98.5, 97.6, and 81.0 percent on the development instances with 15, 45, and 120 jobs. It also lowered total waiting on the three development instances, by 65.6, 61.1, and 61.8 percent, but raised it on the 6-job instance from 13 to 37 minutes. Figure 1 shows both effects by instance size: the tardiness of the searched sequence is smallest relative to the rule at 15 and 45 jobs and rises to 19 percent of the rule's value at 120 jobs.

HiGHS proved the optimal total weighted tardiness to be 233 on the 6-job instance in 0.36 seconds and 21 on the 15-job instance in 85.8 seconds, and iterated greedy reached both values. On the 15-job instance, the solver's optimal order had 323 minutes of waiting and iterated greedy's had 252, because only iterated greedy breaks ties in Z by waiting. With seeds 1 to 5, iterated greedy returned exactly the default seed's tardiness on the 6-, 15-, and 45-job instances, and values from 4,159 to 4,306 on the 120-job instance, all below the default seed's 4,383. Its computation time grew from 0.21 seconds at

Table 2: Deterministic comparison. Each row is one method on one instance, and every schedule is feasible. Seconds are the method's computation time on the laptop of section 5.1. The optimum rows come from the mixed-integer program of section 4.2; on the 15-job instance the optimal tardiness of 21 was reached by an order with 323 minutes of waiting.

Instance (jobs)	Method	Total weighted tardiness	Total waiting	Seconds
Hand-checkable (6)	Earliest due date	281	13	0.000
	NEH_EWDD only	233	37	0.000
	Iterated greedy	233	37	0.21
	Optimum (HiGHS)	233	37	0.36
Small development (15)	Earliest due date	1,417	732	0.000
	NEH_EWDD only	70	291	0.000
	Iterated greedy	21	252	0.52
	Optimum (HiGHS)	21	323	85.8
Medium development (45)	Earliest due date	2,371	2,040	0.000
	NEH_EWDD only	360	1,068	0.000
	Iterated greedy	58	794	3.22
Large development (120)	Earliest due date	23,015	7,863	0.000
	NEH_EWDD only	9,452	3,928	0.003
	Iterated greedy	4,383	3,005	37.2

6 jobs to 37.2 seconds at 120 jobs, while the rule took under a millisecond on every instance.

6.2 The fixed plan keeps the lower tardiness in all 30 scenarios

Table 3 summarizes the two plans for the 45-job instance over the 30 saved scenarios at $\nu = 0.2$. The iterated greedy plan had the lower weighted tardiness in all 30 scenarios, with a mean of 420.6 weighted minutes against 3,541.1, a reduction of 3,120.5 weighted minutes (standard error 150.1 over the paired scenarios) or 88.1 percent, and a 90th percentile of 592.2 against 4,616.6. In every scenario the reduction was at least 1,948.6 weighted minutes, and its 10th percentile was 2,235.1. Its mean waiting was 1,276.9 minutes against 2,454.5, a reduction of 48.0 percent. Neither plan broke a hard constraint in any scenario, because the execution rule preserves release, stage order, and one job per stage for any processing times, and due dates are soft targets. Figure 2 shows that the ranges do not overlap: the iterated greedy plan's weighted tardiness ran from 103.3 to 1,120.0, and the rule's from 2,276.2 to 6,013.8.

Table 3: Stress test on the 45-job instance: 30 scenarios at a coefficient of variation of 0.2, seed 20261103. "Planned" is the plan's value on the exact data; the mean and the 90th percentile are over the scenarios. Tardiness is in weighted minutes and waiting in minutes.

	Total weighted tardiness			Total waiting			Scenarios breaking
Plan	Planned	Mean	90th pct.	Planned	Mean	90th pct.	a hard constraint
Earliest due date	2,371.0	3,541.1	4,616.6	2,040.0	2,454.5	2,768.6	0
Iterated greedy	58.0	420.6	592.2	794.0	1,276.9	1,454.9	0

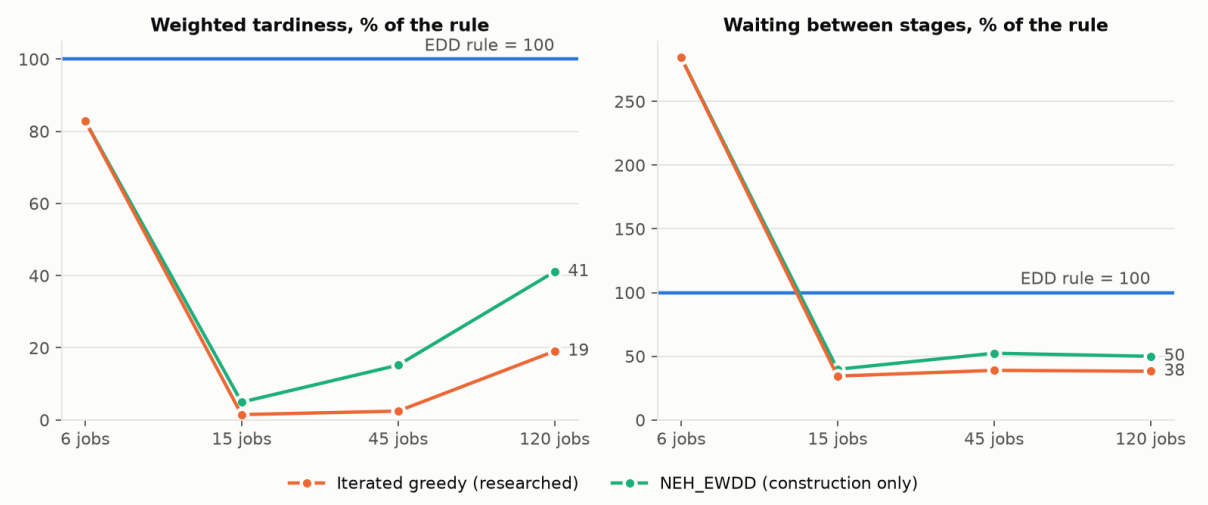


Figure 1: Total weighted tardiness (left) and total waiting (right) of iterated greedy and of NEH_EWDD alone, each as a percentage of the earliest due date rule's value on the same instance (the rule is the line at 100). The horizontal axis gives the instance size; the labels at the right edge are the 120-job values. The two methods coincide on the 6-job instance.

6.3 The relative advantage narrows as variability grows, while the mean absolute advantage stays above 2,500 weighted minutes

Table 4 and fig. 3 repeat the execution at five coefficients of variation, each with 30 new scenarios. The iterated greedy plan's mean weighted tardiness rose from 148.1 at $\nu = 0.1$ to 8,901.6 at $\nu = 0.8$, and the rule plan's from 2,720.1 to 12,230.5. The relative reduction in mean weighted tardiness fell from 94.6 percent at $\nu = 0.1$ to 27.2 percent at $\nu = 0.8$. The mean absolute reduction stayed between 2,572.0 and 3,683.7 weighted minutes at every level: it rose from $\nu = 0.1$ (standard error 65.9) to $\nu = 0.5$ (standard error 223.8), and its fall of 354.8 to 3,329.0 at $\nu = 0.8$ (standard error 342.7) is within the combined standard error of 409.3 of the two estimates. The gain in a single period became less dependable at $\nu = 0.8$: the 10th percentile of the per-scenario reduction, between 1,781.1 and 2,473.2 weighted minutes at the lower levels, fell to 851.5, and the rule's plan won 1 of the 30 scenarios, by 870.4 weighted minutes, its only win in the 180 paired scenarios of the six sets.

Table 4: Sweep over variability on the 45-job instance. Each row uses 30 new scenarios, from seeds 6522 to 6526 in order. Tardiness is in weighted minutes. The reduction is the per-scenario difference between the two plans' weighted tardiness: its mean, with the standard error in parentheses, and its 10th percentile; the percentage compares the two plans' means.

	Earliest due date		Iterated greedy		Reduction		
ν	Mean	90th pct.	Mean	90th pct.	Mean (s.e.)	10th pct.	Percent
0.1	2,720.1	3,171.6	148.1	228.4	2,572.0 (65.9)	2,048.3	94.6
0.2	3,273.5	4,425.0	407.2	643.9	2,866.4 (121.3)	2,055.3	87.6
0.3	4,139.8	6,213.4	952.5	1,676.0	3,187.3 (203.8)	1,781.1	77.0
0.5	6,680.4	9,498.8	2,996.7	5,434.2	3,683.7 (223.8)	2,473.2	55.1
0.8	12,230.5	17,420.0	8,901.6	14,219.8	3,329.0 (342.7)	851.5	27.2

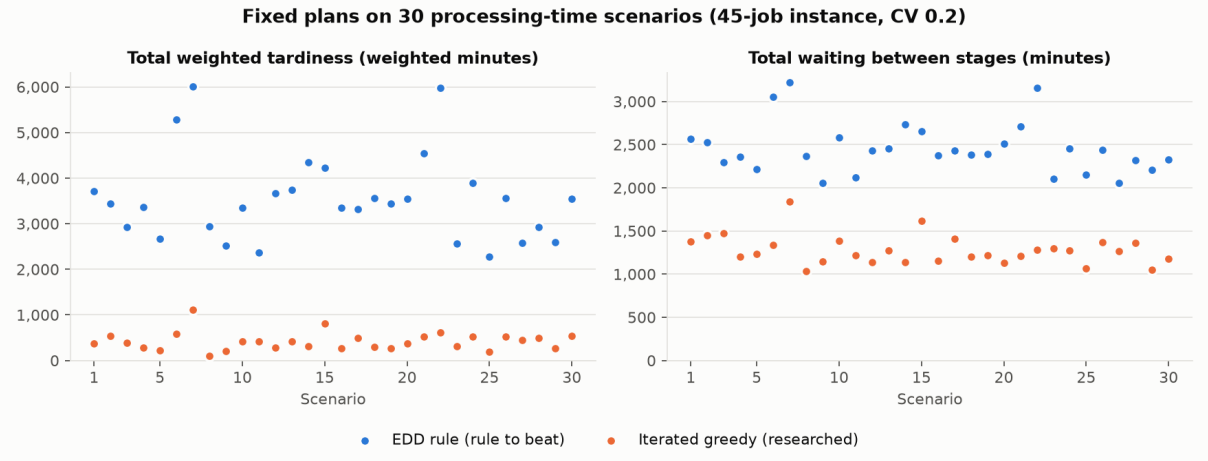


Figure 2: Total weighted tardiness (left) and total waiting (right) of the two fixed plans in each of the 30 scenarios of table 3. Each dot is one plan in one scenario, and both plans face the same realized processing times in a scenario.

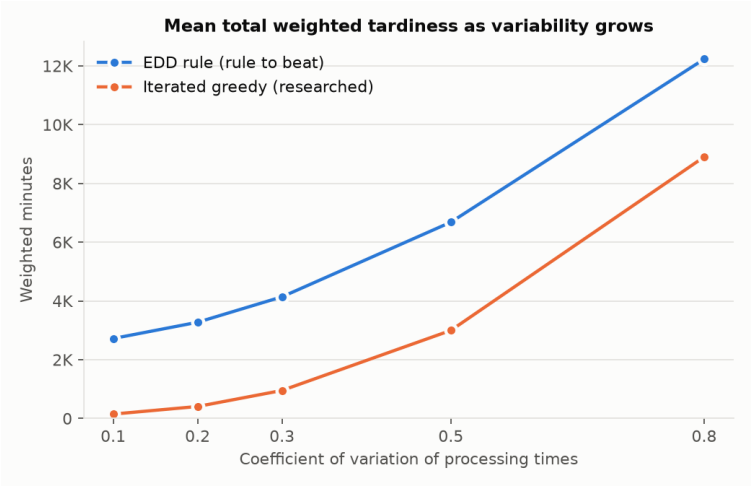


Figure 3: Mean total weighted tardiness of the two fixed plans for the 45-job instance at five coefficients of variation of the processing times, 30 scenarios each (table 4). The vertical gap between the lines is the absolute reduction, between 2,572.0 and 3,683.7 weighted minutes at every level.

7. Findings

7.1 Weights decide which jobs go first

Figure 4 draws both schedules of the 6-job instance. Earliest due date places job J005, due at 67 with weight 1, ahead of jobs J003 and J004, which carry weight 4. Iterated greedy moves J005 to the end, so J003 finishes at 97 and J004 at 119, against 108 and 139 under the rule, while J005 finishes at 136, against 82. The weighted tardiness of J003 and J004 falls by 124 weighted minutes and that of J005 rises by 54, and with a smaller rise for J006 the total falls from 281 to 233. The same reordering raises waiting on this instance from 13 to 37 minutes, because the two jobs now at the end, J006 and J005, wait 18 and 19 minutes at stages S2 to S4 behind the long operations of the weight-4 jobs.

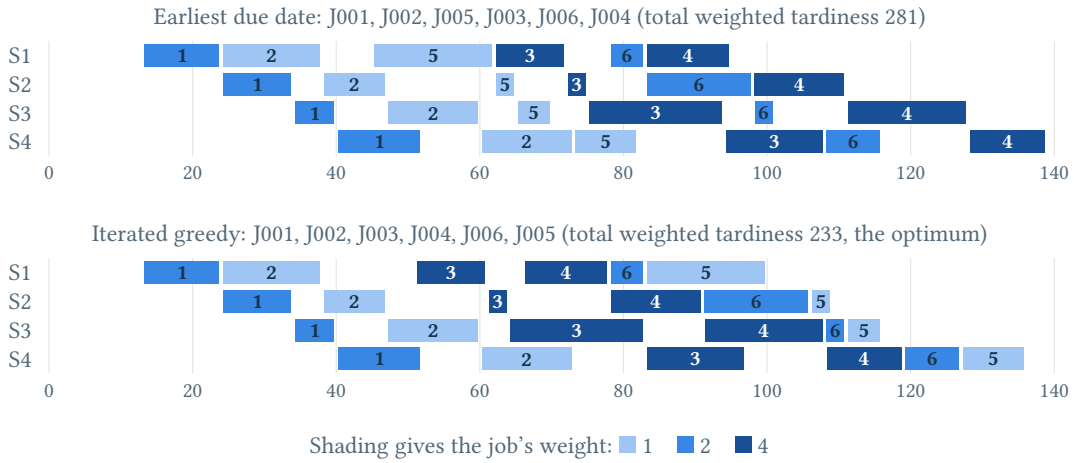


Figure 4: Both schedules of the 6-job instance on stages S1 to S4, with time in minutes on the horizontal axis. Each bar is one operation, labeled with its job number and shaded by the job's weight. Iterated greedy moves the weight-1 job J005 behind the weight-4 jobs J003 and J004.

On the 45-job instance, the share of jobs finishing on time rose from 0.244 under earliest due date to 0.911 under iterated greedy, and on the 15-job and 120-job instances from 0.333 to 0.867 and from 0.233 to 0.642.

7.2 The search removes most of the tardiness the weighted start leaves, and 120 jobs need more of it

NEH_EWDD alone lowered weighted tardiness relative to the rule by 95.1, 84.8, and 58.9 percent on the 15-, 45-, and 120-job instances (table 2). The iterations of Algorithm 1 then lowered NEH_EWDD's value by a further 70.0, 83.9, and 53.6 percent, or 49, 302, and 5,069 weighted minutes. On the 120-job instance the search had not converged after 5000 iterations, since five other seeds found sequences 1.8 to 5.1 percent below the default seed's 4,383. Better sequences are therefore within reach of the same search on the largest instance, at a computation time that already reaches 37.2 seconds.

On the 15-job instance, the solver's optimal order and iterated greedy's order both have tardiness 21, but the solver's has 323 minutes of waiting and iterated greedy's 252 (table 2). Among sequences with optimal tardiness, the tie-break thus removes 71 minutes of work in process that a tardiness-only objective leaves in place.

7.3 Variability makes the searched plan's on-time jobs late, without thinner margins

At $\nu = 0.2$, the iterated greedy plan had 4 late jobs as planned and 11.5 on average across the 30 scenarios, while the rule's plan had 34 and 36.1. The rule's mean weighted tardiness rose by 1,170.1 weighted minutes over its plan while its count of late jobs rose by only 2.1, so most of the increase came from jobs that were already late finishing later still. The iterated greedy plan's mean rose by 362.6, a factor of 7.25 over its planned 58, against a factor of 1.49 for the rule, because its losses come from jobs that were on time in the plan.

The on-time jobs of the iterated greedy plan had a median slack of 31 minutes before their due dates, against 28 for the rule's plan. In each plan, 3 on-time jobs had slack below one standard deviation of their own total processing time at $\nu = 0.2$, which is $0.2\sqrt{\sum_k p_{jk}^2}$ for job j. The iterated greedy plan is

exposed because it has 41 on-time jobs that a delay can make late, against 11 for the rule.

8. Discussion

8.1 The searched sequence saves 2,866.4 to 3,120.5 weighted minutes per period at $\nu = 0.2$, at the cost of a hand-traceable rule

We recommend that the planner sequence the line with iterated greedy. For a period like the 45-job instance with processing times varying at a coefficient of variation of 0.2, its fixed plan lowers mean total weighted tardiness by 3,120.5 weighted minutes (standard error 150.1) on the main scenarios and by 2,866.4 (standard error 121.3) on an independent second set, leaves 11.5 jobs late per period on average against 36.1, and cuts mean waiting between stages by 48.0 percent. The planner gives up a rule that can be traced by hand, accepts a result that depends on a seed and takes seconds to compute, 3.22 at 45 jobs and 37.2 at 120, and on very small batches may accept more waiting, as on the 6-job instance. The choice needs a fresh comparison if processing times vary much more than at $\nu = 0.2$: at $\nu = 0.8$ the mean gain held at 3,329.0 weighted minutes, but the 10th percentile of the per-period gain fell to 851.5 and the rule's plan won 1 of 30 scenarios, so a planner who weighs the worst periods most should repeat the comparison at the line's measured variability.

8.2 The evidence covers one generator, independent noise, and a plan that is never revised

All four instances come from one generator, and the stress test covers only the 45-job instance; the instances have processing times of 2 to 20 minutes and median due-date windows of 154 to 158 minutes, so a line with tighter due dates or more uneven stages may show different margins. Our scenarios draw each of the 360 processing times of the 45-job instance independently, so they do not represent a common cause, such as a slow shift or a degraded machine, that lengthens many operations at once. The planner in our study never revises the sequence, which isolates the value of the plan itself and leaves out any gain from rescheduling when actual times are known. Optimality is proven only up to 15 jobs, and on the 120-job instance other seeds found sequences up to 5.1 percent better, so the gains we report for the largest instance understate what the search can deliver.

8.3 Three extensions follow from these limits

A search that scores each candidate sequence by its mean tardiness over sampled scenarios would target the variability directly and could narrow the relative gap at high ν, which the sweep of section 6.3 can measure. Scenarios with correlated factors, in which one draw lengthens a whole stage or a whole shift, would test whether the absolute gain survives a common cause. Re-running the search at each release with the times already realized would show how much of the stress-test loss a revised plan recovers.

9. Conclusions

On exact data, iterated greedy lowered total weighted tardiness relative to the earliest due date rule by 81.0 to 98.5 percent on the development instances and reached the proven optimum on both instances for which we certified one. Held fixed while processing times varied at a coefficient of variation of 0.2, its plan had the lower weighted tardiness in all 30 scenarios and an 88.1 percent lower mean. As variability rose to a coefficient of 0.8, the relative reduction fell to 27.2 percent and the 10th percentile of the per-period reduction to 851.5 weighted minutes, while the mean absolute reduction stayed between 2,572.0 and 3,683.7 weighted minutes.

For a planner who must fix one order before the work starts, we found a searched sequence whose mean gain over the due-date rule held at every level of processing-time variability we tested, and variability, the one factor we varied, decided how dependably that gain arrived in a single period.

Code and data. The code, the supplied data, every decision, and every result in this report are in the project repository at `https://github.com/mutingma123/OM522Project`, where one command regenerates every table and figure.

References

[1] Eva Vallada, Rubén Ruiz, and Gerardo Minella. Minimising total tardiness in the m-machine flowshop problem: A review and evaluation of heuristics and metaheuristics. *Computers & Operations Research*, 35(4):1350-1373, 2008. doi:10.1016/j.cor.2006.08.016.

[2] Rubén Ruiz and Thomas Stützle. A simple and effective iterated greedy algorithm for the permutation flowshop scheduling problem. *European Journal of Operational Research*, 177(3):2033-2049, 2007. doi:10.1016/j.ejor.2005.12.009.

[3] Rubén Ruiz and Thomas Stützle. An Iterated Greedy heuristic for the sequence dependent setup times flowshop problem with makespan and weighted tardiness objectives. *European Journal of Operational Research*, 187(3):1143-1159, 2008. doi:10.1016/j.ejor.2006.07.029.

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A. Notation

Tables 5 to 7 list every symbol in the report with its meaning, its units, and where it is defined. No symbol carries two meanings.

Table 5: Indices and data.

Symbol	Meaning	Units	Defined in
n	Number of jobs	jobs	section 4.1
m	Number of stages	stages	section 4.1
j	Job index, $j = 1, \ldots, n$		section 4.1
h	Job index in sums and maxima that already involve j		section 4.2
k	Stage index in line order, $k = 1, \ldots, m$		section 4.1
i	Position index within a sequence π		section 4.1
q	Position index in the mixed-integer program		section 4.2
r_j	Release time of job j	minutes	section 4.1
d_j	Due date of job j	minutes	section 4.1
w_j	Weight (priority score) of job j	integer score	section 4.1
p_{jk}	Processing time of job j at stage k	minutes	section 4.1

Table 6: Schedules, measures, and variables of the mixed-integer program.

Symbol	Meaning	Units	Defined in
π, $\pi(i)$	A sequence of all jobs, and the job in its position i		section 4.1
S_{jk}, C_{jk}	Start and completion of job j at stage k	minutes	(4.1)
T_j	Tardiness of job j, $\max\{0, C_{jm} - d_j\}$	minutes	section 4.1
$Z(\pi)$	Total weighted tardiness of π (primary measure)	weighted minutes	(4.2)
$W(\pi)$	Total waiting between stages of π (secondary measure)	minutes	(4.2)
$F(\pi)$	The pair $(Z(\pi), W(\pi))$, compared lexicographically		section 4.3
x_{jq}	1 if job j takes position q, 0 otherwise	binary	section 4.2
B_{qk}, E_{qk}	Start and end of the job in position q at stage k	minutes	section 4.2
U_j	Upper bound on the tardiness of job j in (4.9)	minutes	section 4.2

Table 7: Method and experiment settings.

Symbol	Meaning	Value or units	Defined in
π^{EDD}	Sequence of the earliest due date rule		Algorithm 1
π^*, π', π''	Best sequence found, a rebuilt sequence, and a local search candidate		Algorithm 1
δ	Number of jobs removed in each iteration	4	section 4.3
λ	Temperature factor	0.5	section 4.3
Θ	Temperature, $\lambda \sum_{j,k} p_{jk}/(10\,n\,m)$, compared with increases in Z	minutes	section 4.3
Δ	Increase in Z of a candidate over the current sequence	weighted minutes	section 4.3
L, ℓ	Iteration limit and iteration counter	$L = 5000$	Algorithm 1
u	Uniform random draw on $[0, 1]$ in the acceptance test		Algorithm 1
ω	Scenario index	30 per test	section 5.3
$\varepsilon_{jk\omega}$	Lognormal factor on p_{jk} in scenario ω, mean 1		section 5.3
v	Coefficient of variation of each factor	0.1 to 0.8	section 5.3
σ	Standard deviation of $\ln \varepsilon_{jk\omega}$, $\sigma^2 = \ln(1 + v^2)$		section 5.3